

1. Domain & System Hosting

- An **existing domain** is currently available and can be used to deploy the system through a dedicated **subdomain**.
- This is suitable for the initial implementation and can reduce the initial setup cost.
- However, the **recommended approach is to register a new dedicated domain** for the ERP.
- A dedicated domain provides better scalability and separation from existing systems.
- In the future, the same domain can be used as the **central platform for the entire authority's ERP**, with separate portals/modules for different departments.
- Example structure:
 - `erp-domain.lk` – Main ERP platform
 - `department1.erp-domain.lk` – Department 1
 - `department2.erp-domain.lk` – Department 2
 - `department3.erp-domain.lk` – Department 3
- The hosting infrastructure should be selected so that server resources can be upgraded as the number of users and departments increases.

2. Current User Requirements

- Initial number of users: **approximately 5 users**.
- The system should not be designed only for 5 users, as the ERP is expected to expand in the future.
- The infrastructure should therefore support **horizontal and/or vertical scaling** when additional departments and users are introduced.
- Additional users, modules, and departments can be added without requiring a complete infrastructure replacement.

3. Database – MongoDB

- **MongoDB** will be used as the primary application database.
- Initial expected database size:
 - Approximately **5,000 records**.
 - Each record may contain approximately **2–4 PDF/PNG documents**.
- MongoDB will primarily store:
 - Record information
 - User information
 - Application data
 - Document metadata
 - Secure references/URLs to stored documents
- Actual PDF and PNG files should **not be stored directly inside MongoDB**.
- The database can be scaled to a higher-tier cloud configuration as the ERP grows.

4. Document Storage – AWS

- **Amazon Web Services (AWS)** is recommended for storing PDF and PNG documents.
- **Amazon S3** can be used as the primary document storage service.
- The system will store the actual files in AWS S3 while MongoDB stores the corresponding file references and metadata.
- This approach provides:
 - Scalable storage
 - Secure file access
 - Easier backup and recovery
 - Better database performance
 - Ability to handle a significantly larger number of documents in the future
- AWS S3 storage can be expanded without requiring major changes to the application architecture.
- Access to documents should be controlled through the ERP's authentication and authorization system.

5. Scalability Considerations

- The initial infrastructure will be sized according to the current requirement of approximately **5 users and 5,000 records**.
- The architecture should nevertheless be designed for future expansion of:
 - Users
 - Departments
 - ERP modules
 - Database records
 - PDF/PNG documents
 - File storage requirements
 - System traffic
- Server resources, database capacity, and AWS storage can be increased progressively based on actual usage.
- This avoids paying for a large infrastructure before it is required while maintaining a clear path for future ERP expansion.

6. Rough Cost Estimate

Component	Estimated Monthly Cost	Estimated Annual Cost
New Domain	—	LKR 4,000 – 8,000
Application Hosting / Server	LKR 2,500 – 6,000	LKR 30,000 – 72,000
MongoDB	LKR 0 – 4,000	LKR 0 – 48,000
AWS S3 Document Storage	LKR 500 – 2,000	LKR 6,000 – 24,000
Backup / Additional Cloud Usage	LKR 500 – 2,000	LKR 6,000 – 24,000
Estimated Total	LKR 3,500 – 14,000	LKR 42,000 – 168,000

7. Cost Notes

- The above figures are **rough estimates for the initial ERP deployment** and are not fixed service charges.
- Actual costs will depend on:
 - Number of users
 - Server specifications
 - Database usage
 - Total document storage
 - File download/upload traffic
 - Backup requirements
 - Number of future departments and modules
- As the ERP expands, infrastructure costs will increase accordingly.
- The recommended approach is to **start with an appropriately sized infrastructure and scale it based on actual usage**, rather than purchasing large infrastructure capacity from the beginning.
- If the existing domain is used initially, the **new-domain registration cost can be excluded from the initial deployment cost**.

8. A practical estimate

Item	Initial Recommended Limit
Users	5 concurrent users
Records	~5,000
Files per record	2–4
Maximum file size	10–20 MB/file
Total possible files	10,000–20,000 files
Estimated storage	~ 50–400 GB , depending on file sizes
Daily uploads	Hundreds to thousands
Daily downloads	Hundreds to thousands
Concurrent file operations	5–20+ users/operations comfortably, depending on hosting